

Date: 03/06/2026

NATIONAL SKILL TRAINING INSTITUTE (WOMEN), INDORE

**ARTIFICIAL INTELLIGENCE PROGRAMMING ASSISTANT
(APPLIED AI - INTERNAL PRACTICAL)**

2025-26

TIME: 3 Hrs

MARKS: 50

Note: Attempt any 2

1. Building a Student Performance Prediction System using GitHub

Problem Statement

To creating a machine learning project that predicts whether a student will pass or fail based on study hours, attendance, and previous scores. The project must be managed using GitHub with proper version control.

The task:

1. Create GitHub Repository
2. Load and Explore Dataset
3. Preprocess Data
4. Train ML Model
5. Push Code & Collaborate

2. Car Price Prediction Dashboard

Problem:

Develop an interface to predict used car prices using features like year, fuel type, and mileage.

Task:

1. Load and clean car dataset.
2. Perform regression modeling.
3. Visualize feature importance.
4. Create user input interface.
5. Display predicted price with charts.

3. Predicting Diabetes Risk via REST API

Problem:

Develop a REST API that predicts a person's risk of diabetes based on health parameters such as BMI, age, blood pressure, and glucose levels. The API should accept JSON input and return the predicted risk score or classification.

Task:

1. Train a classification model (Logistic Regression or Random Forest) on a diabetes dataset and save the model.
2. Build a Flask or FastAPI application with a /predict endpoint.
3. Accept JSON input for user health parameters and preprocess it for the model.
4. Return the prediction (risk probability or class) in JSON format.
5. Test the API locally and validate predictions using sample JSON requests.

Task 2: Write an Effective Text Prompt

Create a detailed prompt describing the visual you want to generate.

“A modern AI classroom with students using laptops, futuristic design, bright colors, flat illustration style”

Task 3: Generate and Refine Images

Generate multiple image variations and refine results by:

- Changing styles (illustration, photo, 3D, watercolor)
- Adjusting mood and color tone
- Modifying prompt keywords

Task 4: Customize and Edit the Visual

Select the best-generated image and enhance it using Canva tools:

- Add text, icons, and shapes
- Resize for presentation, poster, or social media
- Apply filters and layout adjustments

Task 5: Export and Evaluate the Output

- Download the final visual and evaluate:
- Prompt relevance
- Visual clarity and creativity
- Suitability for the intended use case